

Lifestyle Intervention: Results of the Treatment of Mild Hypertension Study (TOMHS)¹

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Background. Questions regarding the efficacy of nonpharmacologic approaches for the treatment of stage 1 hypertension were addressed as part of the Treatment of Mild Hypertension Study (TOMHS), a 4-year, randomized clinical trial ($N = 902$). This report describes the lifestyle intervention program used in TOMHS, presents data on the lifestyle changes observed, and focuses on the effect of weight loss on blood pressure and blood lipid levels.

Methods. Participants were randomly assigned to receive either placebo or one of five different antihypertensive medications. All took part in a lifestyle intervention program to reduce weight and sodium and alcohol intake and to increase physical activity.

Results. Substantial changes from baseline levels were achieved for all lifestyle intervention variables. Mean weight change was -10.5 lb (-5.6%) at 1 year, -8.5 lb (-4.5%) at 2 years, -7.4 lb (-4.0%) at 3 years, and -5.7 lb (-3.0%) at 4 years. At 4 years, 70% of participants remained below baseline weight and 34% maintained a weight loss of 10 lb or greater. Mean change in urinary sodium excretion was -12.5 mmol/8 hr (-23%) at 1 year, -10.7 mmol/8 hr (-20%) at 2 years, -8.4 mmol/8 hr (-16%) at 3 years, and -4.6 mmol/8 hr (-9%) at 4 years. Alcohol intake declined by 1.6 drinks/

week among drinkers at 4 years. Reported leisure physical activity increased by 86% at 1 year and remained 50% above baseline at 4 years. Beneficial changes in blood pressure and serum lipids were associated with these changes.

Conclusions. These results support a role for lifestyle interventions as the initial treatment for stage 1 hypertension and demonstrate that such interventions can be successfully implemented in the clinical setting. © 1995 Academic Press, Inc.

INTRODUCTION

Approximately 30% of hypertensives, or roughly 25 million Americans, are considered to be mild hypertensives [diastolic blood pressure (DBP) 90–99 mm Hg, systolic blood pressure (SBP) 140–159 mm Hg] (1). Results of the Hypertension Detection and Follow-Up Program showed a 20% reduction in morbidity and mortality in mild hypertensives treated with stepped-care pharmacologic treatment compared with usual care (2). However, concerns have been raised regarding the appropriateness of long-term drug therapy for mild hypertensives and there is interest in the possibility of treating these patients with nonpharmacologic therapy, either alone or in combination with antihypertensive medication. The Fifth Report of the Joint National Committee on Detection, Evaluation and Treatment of High Blood Pressure (JNC V) (1) and the recent report of the National High Blood Pressure Education Program on the Primary Prevention of Hypertension (3) emphasize the importance of nonpharmacologic treatment approaches such as weight loss, sodium reduction, decreased alcohol intake, and increased physical activity.

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In spite of this consensus regarding the value of non-pharmacologic therapy, two important questions remain: (a) are the recommended lifestyle modifications efficacious in the treatment of mild hypertension and (b) are patients able to achieve and maintain desirable changes in these lifestyle factors? These issues were addressed in the Treatment of Mild Hypertension Study (TOMHS), a multicenter clinical trial in patients with mild hypertension. Detailed descriptions of the methods and the main results of TOMHS have been published previously (4, 5). This report describes the lifestyle intervention program used in TOMHS, presents data on the lifestyle changes observed, and focuses on the effect of weight loss on blood pressure and blood lipid levels. Data on gender differences in response to the intervention are also presented.

METHODS

Design

TOMHS was a 4-year, multicenter, randomized, double-blind, placebo-controlled trial designed to compare six treatments for long-term care of individuals with mild hypertension. Clinical centers were located in Birmingham, Alabama; Chicago, Illinois; Minneapolis, Minnesota; and Pittsburgh, Pennsylvania. Units for statistical coordination, administration, ECG coding, and central laboratory analysis were at the University of Minnesota, Minneapolis, Minnesota. All participants took part in a lifestyle intervention program to reduce weight, decrease sodium and alcohol intake, and increase leisure physical activity. Participants also received either a placebo or one of five antihypertensive medications. For patients whose blood pressure (BP) went above specified levels (DBP $\geq$ 95 mm Hg at three consecutive visits or $\geq$ 105 mm Hg at a single visit), the initial dose was doubled. If the elevated BP persisted, step 2 medication was added to the step 1 regimen. Participants were treated and followed for a minimum of 4 years. The primary outcomes of TOMHS were changes in BP, changes in the percentage of individuals requiring step 2 medication, lifestyle intervention changes, side effects related to treatment, and cardiovascular changes.

Study Population

The study population consisted of 902 black and white men and women, ages 45 to 69 years, who were free of renal or cardiovascular disease at entry into the trial. Participants were eligible for the study if they were not on BP medication at baseline and their DBP averaged 90–99 mm Hg over three visits, or if at the first baseline visit they were on a single antihypertensive drug and after a drug withdrawal period of at least 6 weeks their DBP averaged 85–99 mm Hg over three visits. Study exclusion criteria were body weight more

than 150% of ideal or less than 100% of ideal; consumption of greater than 28 alcoholic drinks/week; diabetes; a history or clinical evidence of cardiovascular disease; angina by Rose questionnaire; ECG evidence of left ventricular hypertrophy; consumption of more than 50% of meals away from home; life threatening illness; or unwillingness to participate in the lifestyle intervention program.

Prior to randomization, participants attended a series of three eligibility visits, during which the nature of the study was fully explained. Those who met the eligibility requirements and gave informed consent were randomized in double-blind fashion to the six treatment groups. Following randomization, participants were seen for clinic visits at 1 and 3 months and then at 3-month intervals through 4 years. All study measurements were obtained at these visits. Participants also attended group and individual lifestyle intervention visits. Participants were randomized from October 1986 through March 1988 and follow-up lasted through March 1992.

Study Measures

Blood pressure was measured in the right arm using random-zero Hawksley sphygmomanometers with the participant seated and at rest for 5 min. The average of two readings at each visit was used for endpoint determination and data analyses. Participants were weighed without shoes on balance scales calibrated on a weekly basis with the use of a standard 22.5-kg weight. Fasting venous blood samples were obtained at baseline and annually for determination of total cholesterol and lipid fractions. Participants completed a detailed medical and demographic questionnaire at baseline. Questionnaires completed at 6 months and 1 year included process evaluation questions related to the lifestyle intervention program. Alcohol intake was estimated from an alcohol history. Physical activity levels were estimated from a 7-day recall of all leisure activity administered by trained interviewers. This method has been shown to yield accurate estimates of mean kcal/day expenditure and provides results that are congruent with direct measures of activity (6, 7). Two timed overnight urine collections were obtained at baseline and at selected follow-up visits to estimate change in sodium and potassium intake. Urine sodium excretion is considered a more valid measure of sodium intake than dietary self-report (8) and estimates of sodium intake derived from overnight urine collections have been shown to correlate well with results from 24-hr collections (9, 10). Overnight urine collections have been employed in many trials because they reduce participant burden compared to 24-hr collections. This may improve compliance with the recommended procedure for collection, resulting in a more valid measure. Collecting over 2 nights helped to reduce the in-

traindividual variation. Nutrient intakes were estimated from 1- and 2-day food records collected at baseline and at 1, 2, and 3 years. Food records were collected by nutritionists who were trained and certified by the Nutrition Coordinating Center (NCC) at the University of Minnesota. The records were reviewed with participants and food models were used to improve estimates of portion size. Food record coding and nutrient calculations were done at the NCC.

Intervention Goals

The goals for the lifestyle intervention program were expressed both as study-wide averages (5% reduction in mean body weight and 25% reduction in mean 8-hr urine sodium excretion) and as individual goals (weight loss of 10 lb or to ideal body weight, whichever was less; sodium intake less than 1,600 mg/day; leisure physical activity to equal 600 kcal/week; alcohol intake not to exceed two drinks/day, with one drink defined as 5 oz wine, 1½ oz spirits, or 12 oz beer).

Intervention Format

The lifestyle intervention program began with a 6-month period of intensive intervention. This initial phase consisted of a series of 13 group and individual sessions conducted by nutritionists who were trained in dietary counseling and the use of behavior modification strategies. Sessions were held weekly or bi-weekly through the first 2 months and then monthly for the remainder of the intensive phase. Group sessions were 60–90 min in length and individual sessions lasted 30–60 min. The size of the groups ranged from 10–20 individuals. Participants received a program manual containing informational handouts and a resource booklet (*Food & Fitness Guide*) that listed sodium, calorie, and fat values for foods. All text-based program materials were written to an eighth-grade reading level. Nutritionists were provided with a study manual of operations that included detailed outlines for conducting the intervention sessions.

Following the initial 6-month period, participants continued to meet individually with study nutritionists at intervals of 6–12 weeks, depending on their progress, for the remainder of the study. At these follow-up visits, which typically lasted 30–60 min, nutritionists provided additional counseling to help participants maintain their new eating and activity behaviors or make further changes, if necessary. Special group programs were offered periodically throughout the follow-up period. Topics included cooking demonstrations, presentations by medical experts, activity outings, and competitions. Participants who gained weight were encouraged to take part in weight loss refresher courses that were offered periodically. Participants also received a quarterly newsletter that featured updates on

study progress, nutrition, and exercise recommendations and recipes.

Nutrition Education

Nutrition education focused on diet changes to reduce intake of fat, calories, and sodium. Participants were given a set of eating guidelines that outlined the necessary changes in simple terms (i.e., decrease use of high-fat foods). Participants kept food diaries and figured daily totals for calorie and sodium intake. The daily totals were compared with study goals for these nutrients (1,600 mg/day for sodium; 1,000–1,200 kcal/day for women, and 1,400–1,600 kcal/day for men). The food diaries also made participants more aware of sources of fat, calories, and sodium in foods. Nutritionists helped participants evaluate their habitual eating patterns on a meal-by-meal basis and suggested alternatives for inappropriate choices. Short-term goals for weight loss were set and periodically reevaluated. Participants also set short-term goals for changes in eating behavior. Nutritionists provided training in food-related skills such as meal planning, food purchasing, food preparation, and selecting from restaurant menus. Information on the principles of good nutrition was included to ensure the overall nutritional quality of participants' diets.

Alcohol

Participants were given information about the effects of alcohol on health, particularly blood pressure. Strategies for reducing alcohol intake, such as substituting nonalcoholic beverages and drinking more slowly, were presented. Participants self-monitored alcohol intake and the results of self-monitoring were discussed at individual counseling sessions.

Physical Activity

An unsupervised, home-based physical activity program was included as a component of the lifestyle intervention to facilitate weight loss and maintenance. The goal of the program was to help participants achieve and maintain an increased activity level, primarily through light- to moderate-intensity leisure activities, such as walking. Participants were provided with a schedule of weekly activity goals that specified frequency and duration of activity. The weekly goals increased gradually until the study goal of 600 kcal/week (approximately 150 min of light- to moderate-intensity activity) was reached. Participants kept track of their daily activities using a 7-day activity diary. Participants also received instruction aimed at furthering their knowledge about exercise. Topics included the benefits of regular physical activity, types of exercise, design of an effective exercise program, exer-

cise safety, selection of proper equipment, and ways to add activity to the daily routine.

Behavioral Strategies

The lifestyle intervention program employed several behavior modification strategies that are derived from learning theory (11) and have been refined through use in previous clinical trials conducted by the investigators. In addition to self-monitoring of food intake and activity (described above), participants were taught to recognize and modify environmental cues that lead to overeating (stimulus control). Cognitive-behavioral strategies were used to help participants improve attitudes that influenced eating and exercise behavior (i.e., dealing with negative thoughts and unrealistic expectations). Social support was encouraged by directing certain program materials at spouses and inviting them to attend sessions. Relapse prevention strategies, such as identification of high-risk situations, were addressed near the end of the initial 6-month period.

Statistical Analysis and Data Presentation

For lifestyle intervention changes, data are presented for the entire TOMHS randomized population of 902 participants and for men and women separately to illustrate gender differences (Fig. 1). In Fig. 1, weight and sodium results are expressed as change from baseline, while alcohol and physical activity are displayed as absolute value over the 4-year follow-up period.

Analyses of the change in BP and blood lipids were limited to the lifestyle only (placebo) group ($N = 234$) to determine the effect of the lifestyle intervention alone on these factors. Changes in BP and lipids were also compared by levels of weight change achieved. In these analyses, average change over 4 years is used and calculated as the average of all follow-up values for a participant minus the baseline value. Multiple regression analyses using weight change as a continuous variable were used to assess the statistical significance of the weight loss effect on these parameters adjusting for the baseline variables, clinical center, and use of antihypertensive medication prior to randomization.

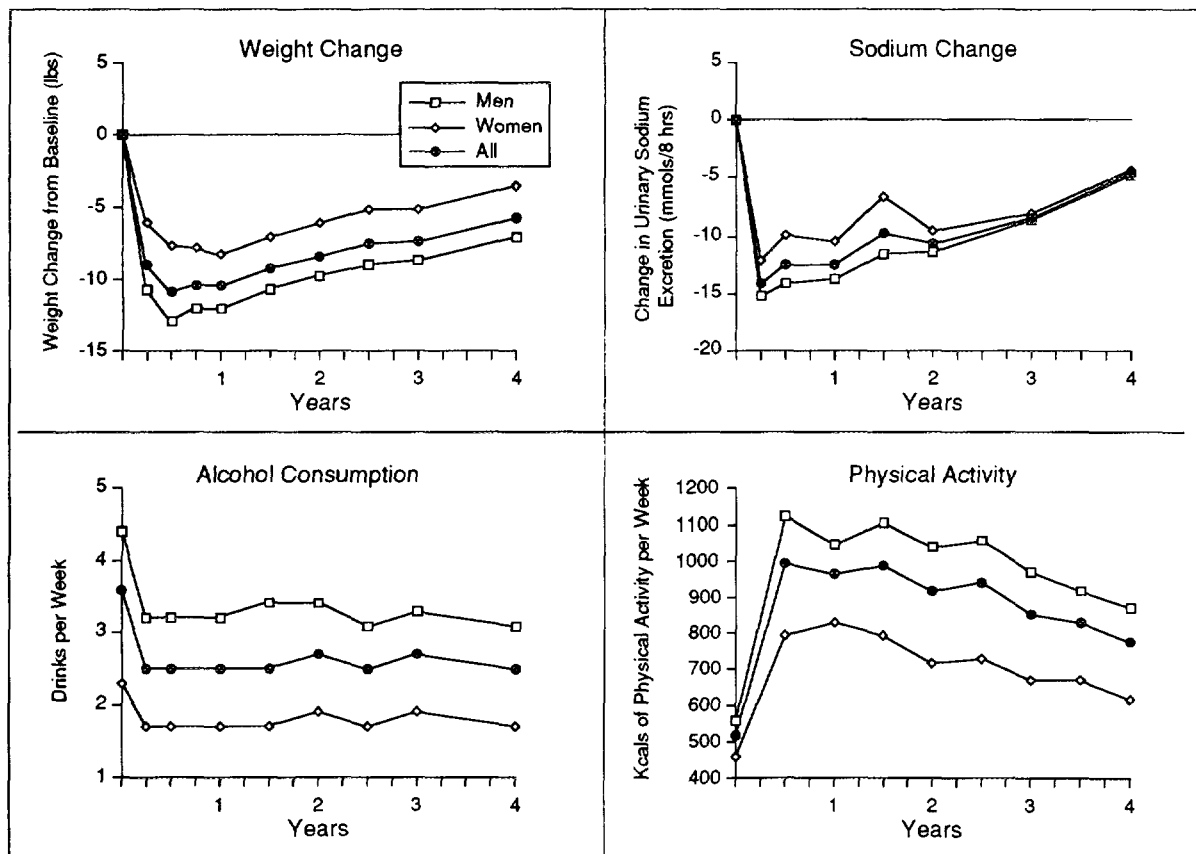


FIG. 1. Lifestyle intervention results over 4 years. Top left, average weight change for all Treatment of Mild Hypertension Study (TOMHS) participants from baseline over 4 years of follow-up. Top right, average change in urinary sodium excretion for TOMHS participants from baseline over 4 years of follow-up. Bottom left, average number of alcoholic drinks consumed per week by all TOMHS participants over 4 years of follow-up (1 drink = 5 oz wine, 12 oz beer, or 1½ oz spirits). Bottom right, average kilocalories of energy expended per week in leisure physical activity by TOMHS participants over 4 years of follow-up.

RESULTS

Participant Characteristics

A total of 902 participants were randomized (234 to the placebo-lifestyle change only group and 668 to treatment groups which combined lifestyle intervention and drug therapy). Baseline demographic and health characteristics are presented in Table 1. Average age at entry was 55 years; participants had a mean of 15 years of education and an average household income of \$39,600 per year. The mean body mass index was 28.9 and the average urine sodium excretion was 53.7 mmol/8 hr (roughly 160 mmol or 3,519 mg of sodium/24 hr). Only 73% reported consuming alcohol at baseline; the average consumption among those who reported consuming alcohol was 5.0 alcoholic drinks/

week. The average leisure physical activity level at baseline was 518 kcal/week; however, 41% of males and 46% of females reported little or no leisure physical activity (less than 160 kcal/week) at baseline.

Attendance

Attendance rates for all aspects of the study were excellent. Attendance at clinic visits averaged 91%. The average number of intervention sessions attended in the first year was 14.2 of a possible 16 (89%). From baseline through 2 years, participants attended 85% of the required intervention visits; during Years 3 and 4 these visits were scheduled based on individual need.

Lifestyle Intervention

Figure 1 shows lifestyle intervention results for the entire TOMHS randomized population and for men and women separately, across all visits. Mean weight loss at the completion of the initial 6-month intervention phase was 10.9 lb, representing a 5.8% reduction from baseline. Most of this initial loss was maintained through 1 year (−10.5 lb or −5.6%) and 2 years (−8.5 lb or −4.5%). At 4 years the overall mean weight loss was 5.7 lb (−3.1%). Thus, more than half of the weight loss achieved initially was maintained through 4 years of follow-up. The distribution of weight change (Fig. 2) shows that after 1 year of follow-up, 89% of participants were below their baseline weight and nearly 50% had lost more than 10 lb. At the end of the 4-year follow-up period, 70% of participants remained below their baseline weight and 34% had maintained at least a 10-lb weight loss.

Figure 1 illustrates that both the amount and rate of weight loss differed between men and women. Men achieved their maximum weight loss of 13.0 lb (−6.5%) at 6 months, while women reached their maximum loss of 8.2 lb (−4.9%) at 1 year. Short-term maintenance of these initial losses was similar for men and women. At 2 years, men had maintained 78% of their initial weight loss while women maintained 73% of their initial loss. Men maintained a greater proportion of their initial loss at 4 years (55 vs 44%).

Significant changes in urinary sodium excretion were observed early in the program and mean sodium excretion remained below baseline levels throughout the trial. The maximum reduction of 14.1 mmol/8 hr (−26%) occurred at 3 months. At 1 year, the mean reduction in overnight urine sodium excretion for all participants was −12.5 mmol/8 hr (−13.7 for men and −10.4 for women). This represents a 23% reduction from baseline. For all participants, percentage reductions from baseline at 2, 3, and 4 years were −20, −16, and −9%, respectively.

At baseline, 73% of the total population consumed alcohol (76% of men and 68% of women). At 4 years, 64% reported consuming alcohol (68% of men and 56%

TABLE 1
Baseline Characteristics of TOMHS Participants

Variable	Men (mean or %)	Women (mean or %)	Total (mean or %)
Demographics			
Age	54.9	54.5	54.8
Male (%)	100.0	0.0	61.8
Black (%)	11.7	32.5	19.6
Married (%)	84.5	64.3	76.8
Education (years)	14.9	14.0	14.6
Annual income (×\$1000)	43.8	32.6	39.6
Employed (%)	72.3	60.8	67.9
Health habits			
Consume alcohol (%)	75.7	67.7	72.7
Mean drinks/week for those consuming alcohol	5.8	3.5	5.0
Smoke cigarettes (%)	9.9	12.5	10.9
Leisure physical activity (kcal/week)	555.4	456.9	517.7
Physiologic measures			
On BP medication at baseline visit 1 (%)	58.2	65.2	60.9
Weight (lb)	199.9	167.2	187.4
Body mass index	29.0	28.6	28.9
Urine sodium excretion (mmol/8 hr)	57.9	46.8	53.7
Urine Na/K ratio	3.7	3.9	3.8
SBP (mm Hg)	139.6	141.5	140.4
DBP (mm Hg)	90.8	90.2	90.5
Cholesterol (mg/dl)	224.8	233.8	228.3
HDL-C (mg/dl)	39.4	50.4	43.6
LDL-C (mg/dl)	156.7	160.9	158.3
Triglycerides (mg/dl)	143.4	112.9	131.7
Number of participants	557	345	902

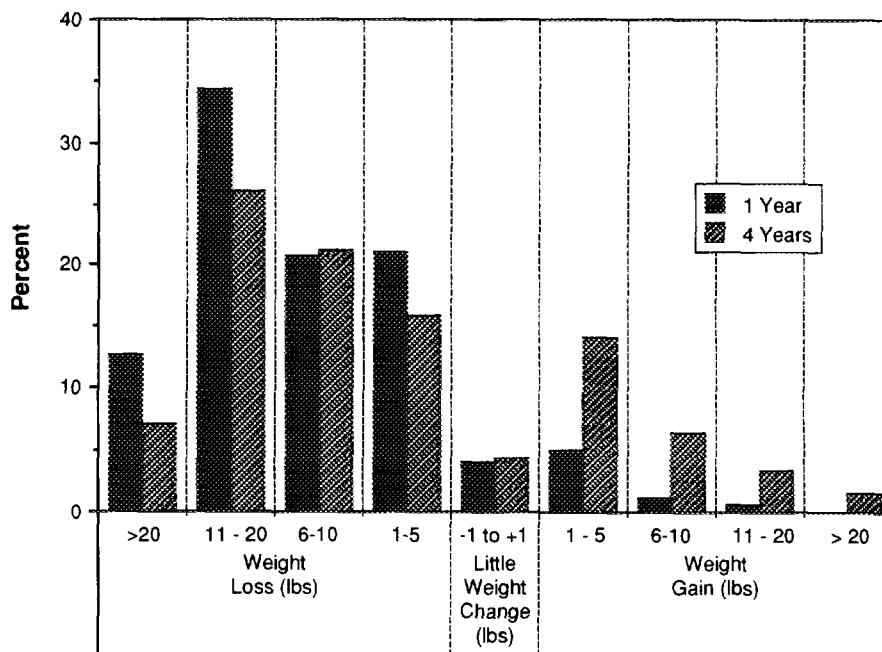


FIG. 2. Distribution of weight change at 1 and 4 years for all TOMHS participants.

of women). Alcohol intake for the whole population decreased by an average of 1.1 drinks/week. Among those reporting consumption of alcohol at baseline, alcohol intake decreased by an average of 1.9 drinks/week for men and 1.1 drinks/week for women, representing a 32% reduction for both groups.

Substantial changes in physical activity were observed during the first year of the intervention, with reported energy expenditure from leisure physical activity increasing by 86% over baseline levels. At 1 year, men reported an expenditure of 1,048 kcal/week (89% above baseline). Women reported expending 827 kcal/week at 1 year (81% above baseline). Walking was the activity most frequently reported, with 50% of participants reporting walking at a moderate pace and 25% reporting walking at a fast pace. These two walking categories accounted for more than half of the kilocalories expended in leisure physical activity at 1 year. The initial increases in leisure physical activity were largely maintained through 2 years. At 4 years, men and women reported leisure physical activity levels 57% and 36% above their baseline levels, respectively.

For men, self-reported daily calorie intake, as calculated from 2-day food records, declined from 2,361 at baseline to 1,770 at 1 year (a 25% reduction). For women, daily calorie intake declined from 1,765 at baseline to 1,360 at 1 year, a reduction of 23%. Percentage of calories from fat decreased from 37 to 30% for men and from 36 to 30% for women at 1 year. For both groups the percentage of calories from saturated fat was 10% at 1 year and 3 years, compared to 13% at baseline. Total dietary sodium intake at baseline was 3,735 mg/day for men and 2,755 mg/day for women. At

1 year, sodium intake had decreased to 1,906 and 1,673 mg/day for men and women, respectively. These changes in reported sodium intake were consistent in direction with the changes observed in urine sodium excretion over the same period. At 3 years, reported sodium intake increased to 2,284 mg/day for men and 1,856 mg/day for women. Intakes of potassium and calcium declined only slightly during the trial, consistent with the caloric reduction. Dietary fiber intake did not change during the study.

Blood Pressure and Blood Lipids

Lifestyle intervention was associated with reductions in DBP and SBP throughout the study (Table 2). At 1 year, in the placebo-lifestyle change only group, DBP had fallen 8.1 mm Hg and SBP 10.6 mm Hg from baseline levels of 90.5 and 141.1, respectively. Reductions in BP were similar at 4 years. Men and women experienced similar changes in BP throughout the study. A substantial number of participants in the placebo-lifestyle change only group were able to achieve BP control without the addition of antihypertensive medication. The proportion of the placebo group remaining at step 1 of the TOMHS protocol (placebo alone) was 81% at 1 year, 74% at 2 years, 70% at 3 years, and 59% at 4 years.

Changes in BP during the trial were associated with weight loss (Table 3). When weight change averaged over all visits is divided into categories (weight gain, lost 1–10 lb, or lost greater than 10 lb) DBP and SBP fall in a linear fashion, with a slightly larger impact on SBP. The trend in BP change by weight change was highly significant ($P < 0.001$) for both DBP and SBP.

TABLE 2

Mean Change in Blood Pressure and Blood Lipids at 1 and 4 Years for TOMHS Lifestyle Intervention Only Participants

	Male		Female		Total	
	1 year mean (SE)	4 years mean (SE)	1 year mean (SE)	4 years mean (SE)	1 year mean (SE)	4 years mean (SE)
Diastolic BP (mm Hg)	-8.3 (0.6)	-8.5 (0.8)	-7.9 (0.9)	-8.7 (0.9)	-8.1 (0.5)	-8.6 (0.6)
Systolic BP (mm Hg)	-10.3 (1.2)	-8.2 (1.3)	-11.0 (1.7)	-9.2 (2.0)	-10.6 (1.0)	-8.6 (1.1)
Total cholesterol (mg/dl)	-7.0 (2.0)	-8.9 (2.4)	-2.4 (2.6)	-1.3 (2.8)	-5.2 (1.6)	-6.0 (1.9)
LDL cholesterol (mg/dl)	-2.3 (1.9)	-6.6 (2.0)	-1.7 (2.3)	-2.7 (2.4)	-2.0 (1.4)	-5.1 (1.5)
HDL cholesterol (mg/dl)	1.5 (0.6)	-0.3 (0.7)	1.7 (0.7)	-0.3 (0.8)	1.6 (0.4)	-0.3 (0.5)
Triglycerides (mg/dl)	-31.3 (5.0)	-9.9 (6.7)	-12.1 (3.8)	8.7 (5.2)	-23.7 (3.4)	-2.8 (4.6)
Number of participants	133	128	88	79	221	207

Important changes in serum lipids were also associated with the lifestyle interventions. As shown in Table 2, total cholesterol (TC) declined from the baseline level of 228 mg/dl by 5.2 mg/dl (-2.3%) at 1 year for the placebo group (-7.0 mg/dl for men and -2.4 mg/dl for women). At 4 years, TC levels remained below the baseline level, with an overall reduction of 6.0 mg/dl. LDL cholesterol (LDL-C) and triglycerides (TG) were also reduced throughout the trial. HDL cholesterol (HDL-C) levels had risen by 1.6 mg/dl at 1 year, but showed a decline of 0.3 mg/dl at 4 years. When considering lipid changes over the entire trial, men experienced greater changes in TC and HDL than women.

Lipid changes also were associated with weight loss, with greater weight losses resulting in larger lipid changes (Table 3). The effect of weight change on lipid levels was statistically significant for TC ($P < 0.001$), LDL-C ($P = 0.005$), HDL-C ($P = 0.005$), and TG ($P < 0.001$). Those who lost more than 10 lb (averaged

across all follow-up visits) showed reductions in TC (-12.3 mg/dl), LDL-C (-8.1 mg/dl), and TG (-31.0 mg/dl). HDL-C for this group increased by 2.0 mg/dl (changes calculated as the average of all follow-up visits - baseline).

Process Evaluation of the Lifestyle Intervention Program

At 6 months, participants were asked to rate the helpfulness of several elements of the lifestyle intervention program using a 4-point scale (not at all helpful to very helpful). With respect to their efforts to lose weight, 91% of participants rated the individual meetings with nutritionists as being either moderately or very helpful. The group meetings were rated as moderately or very helpful by 82% of participants. Use of food diaries was rated as moderately or very helpful by 82%. For sodium reduction, the comparable ratings

TABLE 3

Mean Change over 4 Years in Blood Pressure and Blood Lipids^a by Weight Change for TOMHS Lifestyle Intervention Only Participants

	Weight change category ^b			Regression model	
	Weight gain (n = 48)	Lost 1-10 lb (n = 106)	Lost >10 lb (n = 80)	β Coeff ^c	P value
DBP (mm Hg)	-6.8	-7.8	-10.8	0.2092	<0.001
SBP (mm Hg)	-3.0	-8.8	-13.2	0.4053	<0.001
Total cholesterol (mg/dl)	0.2	-2.0	-12.3	0.6046	<0.001
LDL cholesterol (mg/dl)	-0.6	-1.6	-8.1	0.3724	0.005
HDL cholesterol (mg/dl)	-1.5	2.3	2.0	-0.1273	0.005
Triglycerides (mg/dl)	11.0	-13.5	-31.0	1.7762	<0.001
Mean weight change for category (lb)	3.1	-5.1	-17.0		

^a Mean change over 4 years is calculated as (average of all follow-up visits) - baseline.

^b Weight change is calculated as (average of all follow-up visits) - baseline.

^c Multiple regression on weight change (as a continuous variable), adjusting for baseline levels of the dependent variable and the stratification variables clinical center and use of antihypertensive medication prior to randomization.

were 94% for individual meetings, 90% for group meetings, and 88% for food diaries. For physical activity, the comparable ratings were 73% for individual meetings, 66% for group meetings, and 61% for activity diaries. For all elements and all components, there were no appreciable differences between the ratings provided by men and women.

At 1 year, participants were asked to rate the same elements as to their effect on maintenance of lifestyle changes. A 5-point scale, ranging from negative effect to positive effect, was used. Ninety-two percent of participants saw individual meetings as having had a somewhat positive or positive effect on their ability to maintain their weight loss. Group meetings were seen as having had a somewhat positive or positive effect by 89%, whereas food diaries were seen as having had a somewhat positive or positive effect by 76%. For maintenance of sodium reduction, the comparable ratings were 92% for individual meetings, 89% for group meetings, and 78% for food diaries. For maintenance of physical activity, the comparable ratings were 86% for individual meetings, 81% for group meetings, and 68% for activity diaries. Again, the ratings did not vary appreciably by gender.

DISCUSSION

The 1993 fifth Joint National Committee Report on High Blood Pressure Detection, Treatment and Control recommends lifestyle–nonpharmacologic therapy emphasizing weight loss, sodium reduction, decreased alcohol intake, and increased physical activity as the initial treatment for stage 1 (“mild”) hypertension (1). The report also recommends that lifestyle–nonpharmacologic therapy be combined with antihypertensive drugs for those requiring medication for BP control. However, it is not clear which educational approaches can best help individuals achieve sufficient lifestyle changes for BP control. Also, little is known about the efficacy of programs that intervene on multiple lifestyle factors to maximize BP lowering effect. Finally, the ability of patients to maintain lifestyle changes is a major concern in clinical practice. Results from the TOMHS nonpharmacologic intervention program help to shed light on these issues.

TOMHS participants lost an average of 10.5 lb at 1 year, which is similar to results seen in other blood pressure trials that employed multicomponent lifestyle interventions. After 1 year of intervention, hypertensive patients in the Hypertension Control Program (HCP) (12) lost 8.6 lb, while borderline hypertensives in the Primary Prevention of Hypertension Study (13) lost 8.5 lb. Patients in the Hypertension Intervention Trial lost 6.0 lb after 1 year (14). The 1-year weight loss reported by Applegate et al. (15) for a group of older mild hypertensives in a multicomponent program was 4.6 lb.

TOMHS weight loss results also compare favorably to BP trials in which weight loss was the only intervention component. Mean 1-year weight loss for hypertensives in the Dietary Intervention Study of Hypertension was 7.8 lb (16). In the Hypertension Prevention Trial (17, 18) the 1-year weight loss was 10.9 lb. The 1-year weight loss in the Trials of Hypertension Prevention was 11.0 lb (19). Thus, the results of TOMHS indicate that initial weight loss results do not necessarily suffer within the context of a multicomponent lifestyle intervention program.

Maintenance of weight loss is also an important criterion of program success. TOMHS participants maintained a substantial part of the weight loss achieved during the intensive treatment phase. At 1 year, 90% of participants were below their baseline weight. Participants maintained 78% of their initial weight loss after 2 years and 52% of their initial weight loss through 4 years of follow-up. In comparison, HCP participants maintained all of their initial weight loss through 2 years and 55% of the initial loss through 4 years (12). After 4 years, 70% of TOMHS participants were below their baseline weight and more than 30% had maintained a weight loss of 10 lb or more. Although TOMHS participants did not maintain all of their initial weight loss, the rate of recidivism seen was less than that of other similar trials. This suggests that an approach similar to that used in TOMHS may yield longer lasting benefits, particularly if one assumes that a weight gain of 1–2 lb per year may be the secular norm (20).

Program components that have been linked with greater weight loss include length of follow-up, number and frequency of contacts, a stringent diet prescription, and increased physical activity (21–27). The TOMHS program employed frequent contacts during the first 6 months followed by contacts every 3 months for the remainder of the study. Participants were given a calorie goal and self-monitored their calorie intake frequently during the intensive phase of the intervention and periodically thereafter. Physical activity was a major program component. It is likely that these techniques and the visit frequency contributed to substantial maintenance of the initial weight loss seen throughout the TOMHS program.

The level of sodium reduction reported in BP trials has varied widely. Earlier short-term studies (4–20 weeks) showed sodium reductions of 40 to 50%, as measured by 24-hr urine collections (28–30). More recent trials of longer duration have reported sodium reductions ranging from 13 to 47% (31–41). The trend toward smaller reductions may be due in part to society-wide increases in the frequency of restaurant meals and the use of high-sodium convenience foods. Also, it may be more difficult to maintain sodium reduction over the long term, as there is no convenient way for patients to assess their progress. TOMHS participants

achieved a 26% reduction in sodium excretion by 3 months, slightly exceeding the study goal of a 25% reduction. However, by the end of the trial, urine excretion was only 9% below baseline levels. A similar pattern of recidivism has been reported by other investigators (12–14, 16, 18). Only a few studies have reported sodium results beyond 2 years. Male hypertensive patients in the Minneapolis Mount Sinai Hypertension Trial (a sodium-only intervention) maintained a 25% sodium reduction for 2 years (35). HCP participants maintained a 35% reduction for 4 years (12). The variability in sodium reduction across trials precludes a determination of whether sodium-alone interventions produced greater sodium changes than multicomponent interventions. Despite the increased availability of sodium-modified foods, data from TOMHS and other BP trials suggest that it is becoming more difficult to achieve large reductions in sodium intake in the current food and social environment (16, 36–44).

Increased physical activity contributes to initial weight loss, weight maintenance, and BP lowering (3, 26, 45–47). Physical activity results from TOMHS compare favorably with results from other trials that have incorporated physical activity into multicomponent interventions aimed at improving BP or other indicators of coronary heart disease risk status (12–16, 19). In the Primary Prevention of Hypertension Study (13), 75% of participants reported an unspecified increase in moderate physical activity. Applegate *et al.* (15) used a non-pharmacologic approach that included increased physical activity to treat a group of older mild hypertensives. Participants reported an average increase of 75 to 112 min of moderate physical activity per week, an increase of 50 to 87 min over baseline. In comparison, TOMHS participants reported an average increase above their baseline levels equal to 118 min of moderate physical activity per week at 6 months. The Lifestyle Heart Trial (48) evaluated the effects of a multicomponent intervention including physical activity on atherosclerotic plaque regression. After 1 year, experimental subjects reported an average of 267 min of moderate physical activity per week, an increase of 190 min over baseline. The physical activity goal in this study was slightly higher than the TOMHS goal, at 180 min of moderate activity per week.

Exercise trials and cardiovascular disease rehabilitation programs have generally reported poor success with long-term maintenance of increased physical activity (49, 50). At 1 year, 70% of TOMHS participants were above baseline levels for physical activity and 56% remained above baseline at 4 years.

In contrast to most exercise trials, the TOMHS physical activity program focused on moderate-intensity activities and specified conservative increases in exercise frequency and duration. The program was unsupervised and did not require attendance at an exercise

facility. Behavioral components, such as frequent self-monitoring, were included to enhance compliance. The positive results seen in TOMHS suggest that this approach may be well-suited for use in clinical settings, particularly when patients make repeated clinic visits for BP monitoring.

Responses to the process evaluation questions asked at 6 months and 1 year provide insight into participants' perceptions of the helpfulness of the various intervention approaches used. For the weight, sodium, and physical activity components, the majority of participants found both the individual meetings with nutritionists and the group meetings to be helpful. The use of self-monitoring records, which is perceived as burdensome by many practitioners, was also seen as helpful by the majority of participants. Helpfulness ratings were generally lower for the physical activity component, in spite of the substantial increases achieved by many participants. This may indicate that participants felt capable of increasing activity on their own, or that they perceived didactic presentations to be less relevant in the case of physical activity. The finding that men and women showed no appreciable differences in their perceptions of helpfulness suggests that any of the approaches are equally appropriate for either group.

TOMHS was designed to evaluate the effect of the various treatments on BP control and other physiologic/medical outcomes. The study was limited in the respect that it did not have a no-intervention control group. Thus, conclusions about the efficacy of the lifestyle intervention as a treatment for mild hypertension must be viewed within that context. Also, like any clinical trial, the study population was select. However, TOMHS did recruit from a variety of medical practice settings and included more than 900 people in four distinct geographic locations. Finally, the study was not a demonstration project so conclusions cannot be drawn about the cost-effectiveness of the lifestyle intervention program. In spite of these limitations, it is felt that the study provides valuable information as to what can be achieved when a lifestyle intervention program is conducted within the clinical context, in a systematic and rigorous fashion, and with regular, extended follow-up. Further research is needed to evaluate the cost-effectiveness of lifestyle interventions for BP control. Also, the relative effectiveness of the various intervention delivery approaches and theoretical frameworks should be more fully investigated.

In summary, substantial changes in all intervention variables were achieved during the initial intensive intervention phase of TOMHS and these changes were maintained to a significant degree over 4 years of follow-up. Lifestyle changes were associated with substantial BP lowering and 59% of participants who received placebo along with the lifestyle intervention did not require BP medications over 4 years. Beneficial

changes were seen for other cardiovascular risk factors as well. These results support an important role for lifestyle interventions as the initial treatment for borderline and stage 1 hypertension and as an adjunct to treatment with antihypertensive agents for the control of hypertension.

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